Data in Brief: Hourly Wind Energy Production Dataset from Eskom with Conformal Prediction and Quantile Regression Averaging

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Abstract

In this study, a probabilistic wind energy forecast based on quantile regression averaging is considered within the framework of conformal prediction using Eskom (South Africa) data from April 2019 to November 2023. The partial linear additive quantile regression (PLAQR) model averages forecasts from two models – XGBoost and PCA-based regression. To train and test the model, two datasets are used, with training/validation/testing ratios of 80/10/10 and 85/10/5. Experimental evaluation shows that PLAQR outperforms any individual method, achieving the lowest RMSE and highest directional accuracy.

Specifications Table

Table 1: Specifications Table.

Subject area	Energy Forecasting, Machine Learning
More specific subject area	Short-term Wind Energy Forecasting, Probabilistic Forecasting, Uncertainty Quantification
Type of data	CSV file (hourly time series)
How data were acquired	Publicly downloaded from Eskom Data Portal (https://www.eskom.co.za/dataportal/)
Data format	Raw (engineered features) and preprocessed
Experimental factors	Time series with lagged differences (1, 2, 12, 24 hours), hour-of-day, day-of-week, nonlinear trend component
Experimental features	Preprocessing: STL decomposition, lag feature creation; Modelling: XGBoost, PCR, PLAQR ensemble; Evaluation: RMSE, MAE, MASE, MBE, POCID, fitness metric; Uncertainty: conformal prediction (PICP, MPIW, CWC, PIT histograms)
Data source location	South Africa (Eskom national grid, wind farms aggregated)
Data accessibility	Publicly available at https://github.com/csigauke/Enhancing-Short-Term-Wind-Energy-Forecasting-with-XGBoost-and-Co
Related research article	Nthangeni, R.I., Sigauke, C., Ravele, T., & Tshisikhawe, T.H. (2026). Enhancing Short-Term Wind Energy Forecasting with XGBoost and Conformal Prediction for Robust Uncertainty Quantification. *Computation* (https://doi.org/10.3390/computation14030056).

Value of the Data

- Real-world data for short-term (hourly) energy forecasting based on renewable energy sources, allowing for comparison of tree-based (XGBoost), linear (PCR), and hybrid (PLAQR) approaches.
- Designed features (lagged differences, hour/day, STL-trend) provide a possibility to examine various approaches to feature generation that can help capture daily and seasonal trends.
- A four-year period of the time series provides enough variation to examine how models behave in different weather conditions, thus ensuring validation.
- Prediction confidence intervals for each forecast (with nominal coverage 95%) are included, providing a valuable resource for probabilistic forecasting and grid management, specifically allowing exploration of issues of model sharpness and reliability using metrics such as PICP, MPIW, and PIT histograms.
- PLAQR algorithm itself, as well as the fitness measure that takes into account direction and RMSE together, provides a good example of how to create a combined approach and evaluate its performance.

1 Data

1.1 Dataset Description

The Eskom wind energy dataset contains **5,960 hourly observations** from 02 April 2019 to 28 November 2023. Each record represents the aggregated wind energy production (in MWh) across Eskom's wind farms. The dataset includes **8 variables** capturing temporal dynamics, lagged effects, and nonlinear trends.

1.2 Target Variable

The continuous target variable `WindEnergy` (denoted Y_t) is the hourly wind energy production measured in MWh.

- Minimum: 19.8 MWh
- Maximum: 3102.2 MWh
- Mean: 982.5 MWh
- Median: 903.0 MWh
- Skewness: 0.76 (moderate right skew)
- Kurtosis: 3.31 (leptokurtic, indicating occasional extreme production events)

1.3 Predictor Variables

Table 2: Predictor variables in the Eskom wind energy dataset.

Variable	**Description**	**Type**
difLag1	First-order hourly difference: $Y_t - Y_{t-1}$	Numerical
difLag2	Second-order hourly difference: $Y_t - Y_{t-2}$	Numerical
difLag12	Half-day seasonal difference: $Y_t - Y_{t-12}$	Numerical
difLag24	Daily seasonal difference: $Y_t - Y_{t-24}$	Numerical
Hour	Hour of the day (0–23)	Categorical (encoded)
Day	Day of the week (0–6, Monday=0)	Categorical (encoded)
noltrend	Nonlinear trend component extracted via STL decomposition	Numerical

1.4 Sample Records

Table 3: Sample records from the Eskom wind energy dataset.

WindEnergy	difLag1	difLag2	difLag12	difLag24	Hour	Day	noltrend
1052.3	25.6	51.2	-78.3	-145.2	14	2	890.4
987.6	-64.7	-39.1	-112.5	-201.8	15	2	892.1
1210.5	222.9	158.2	45.2	-33.7	10	3	894.6
875.4	-335.1	-112.2	-98.4	-210.3	22	5	896.2
1345.8	470.4	135.3	124.9	12.6	12	1	898.5

1.5 Performance Metrics

Table 4: Performance metrics used in the analysis.

Metric	Formula
RMSE	$\sqrt{\frac{1}{m}\sum_{t=1}^{m}(y_t - \hat{y}_t)^2}$
MAE	$\frac{1}{m}\sum_{t=1}^{m}\lvert y_t - \hat{y}_t\rvert$
MASE	$\frac{\frac{1}{m}\sum \lvert y_t - \hat{y}_t\rvert}{\frac{1}{n-1}\sum_{t=2}^{n}\lvert y_t - y_{t-1}\rvert}$
MBE	$\frac{1}{m}\sum_{t=1}^{m}(y_t - \hat{y}_t)$
POCID	$\frac{100}{m}\sum_{t=1}^{m} D_t$, $D_t = \mathbf{1}[(y_t - y_{t-1})(\hat{y}_t - \hat{y}_{t-1}) > 0]$
PICP	$\frac{1}{m}\sum_{t=1}^{m}\mathbf{1}[y_t \in [L_t, U_t]]$
MPIW	$\frac{1}{m}\sum_{t=1}^{m}(U_t - L_t)$
CWC	$\text{MPIW} \cdot (1 + \gamma e^{-\eta(\text{PICP}-\mu)})$, $\gamma = 1$ if PICP $< \mu$ else 0

1.6 Software and Packages

Table 5: Software and packages used.

Purpose	Software/Package
Data manipulation	R
Visualisation	ggplot2
Machine learning	pls (PCR), xgboost (XGBoost)
Probabilistic forecasting	Conformal prediction (custom split conformal)
Ensemble method	PLAQR (Partial Linear Additive Quantile Regression)
Programming language	R

2 Results Summary

Table 6: Model performance comparison for two data splits (80/10/10 and 85/10/5).

Split	Model	RMSE	MAE	MASE	POCID (%)	Fitness
80/10/10	XGBoost	182.44	144.15	1.3284	70.71	33.76
	PCR	189.03	149.98	1.3822	71.71	33.60
	PLAQR	**179.25**	**140.99**	**1.2993**	71.15	**34.28**
85/10/5	XGBoost	182.78	143.51	1.2551	72.97	34.80
	PCR	190.21	151.20	1.3224	73.31	34.24
	PLAQR	**178.89**	**139.12**	**1.2168**	**74.24**	**35.81**

Table 7: Conformal prediction interval evaluation (nominal coverage 95%).

Split	Model	PICP	MPIW	CWC
80/10/10	XGBoost	0.9275	654.76	2669.33
85/10/5	XGBoost	0.9364	693.34	2064.10

2.1 Key Findings

1. **PLAQR ensemble** consistently outperforms individual XGBoost and PCR models across all error metrics (RMSE, MAE, MASE) and achieves the highest fitness score, demonstrating the benefit of combining tree-based nonlinearity with linear components.

2. **Conformal prediction** provides well-calibrated prediction intervals: the 85/10/5 split achieves 93.6% coverage (nominal 95%) with an average interval width of 693 MWh. Probability Integral Transform (PIT) histograms confirm uniform distribution of p-values, indicating proper probabilistic calibration.

3. **Directional accuracy** is high across all models (70–74%), with the PLAQR model achieving the best POCID (74.24%) in the larger training set, making it suitable for operational decision-making (e.g., reserve allocation).

4. **Data volume matters**: increasing the training set from 80% to 85% improves all metrics (e.g., PLAQR RMSE reduces from 179.25 to 178.89, POCID increases from 71.15% to 74.24%), confirming that these models benefit from additional historical data.

5. **Variable importance** from XGBoost identifies `noltrend` (nonlinear trend) as the most influential predictor (importance ≈ 0.67), followed by `difLag12` (half-day difference, ≈ 0.23). `Hour` and `difLag24` contribute moderately, while `Day` has negligible impact.

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